Q1 2021 (01 Sep Shift 2)

Let X be a random variable with distribution.

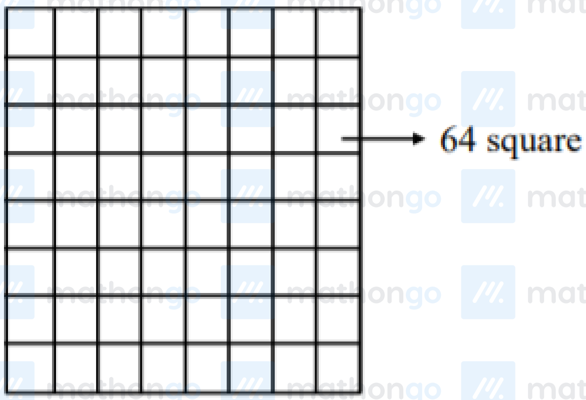
x	–2	–1	3	4	6
P(X = x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2,

then $100\sigma^2$ is equal to :

Q2 2021 (01 Sep Shift 2)

Two squares are chosen at random on a chessboard (see figure). The probability that they have a side in common is :



(1) $\frac{2}{7}$
(2) $\frac{1}{18}$
(3) $\frac{1}{7}$
(4) $\frac{1}{9}$

Q3 2021 (31 Aug Shift 1)

An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p, then 98p is equal to .

Q4 2021 (27 Aug Shift 2)

The probability distribution of random variable X is given by:

X	1	2	3	4	5
P(X)	K	2K	2K	3K	K

Let $\mathrm{p}=\mathrm{P}(1<\mathrm{X}<4 \mid \mathrm{X}<3)$. If $5\mathrm{p}=\lambda\mathrm{K}$, then λ equal to .

Q5 2021 (27 Aug Shift 2)

Each of the persons A and B independently tosses three fair coins. The probability that both of them get the same number of heads is :

(1) $\frac{1}{8}$

(2) $\frac{5}{8}$

(3) $\frac{5}{16}$

(4) 1

Q6 2021 (27 Aug Shift 1)

When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6}-\mathrm{x}$ and its opposite face occurs with probability $\frac{1}{6}+x$. All other faces occur with probability $\frac{1}{6}$. Note that opposite faces sum to 7 in any die. If $0<x<\frac{1}{6}$, and the probability of obtaining total sum $=7$, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is:

(1) $\frac{1}{16}$

(2) $\frac{1}{8}$

(3) $\frac{1}{9}$

(4) $\frac{1}{12}$

Q7 2021 (26 Aug Shift 2)

A fair die is tossed until six is obtained on it. Let X be the number of required tosses, then the conditional probability $\mathrm{P}(\mathrm{X} \geq 5 \mid \mathrm{X} > 2)$ is :

(1) $\frac{125}{216}$

(2) $\frac{11}{36}$

(3) $\frac{5}{6}$

(4) $\frac{25}{36}$

Q8 2021 (26 Aug Shift 1)

Let A and B be independent events such that $\mathrm{P}(\mathrm{A}) = \mathrm{p}, \mathrm{P}(\mathrm{B}) = 2\mathrm{p}$. The largest value of p, for which P (exactly one of A, B occurs)$= \frac{5}{9}$, is :

(1) $\frac{1}{3}$

(2) $\frac{2}{9}$

(3) $\frac{4}{9}$

(4) $\frac{5}{12}$

Answer Key

Q1 (781)	**Q2** (2)	**Q3** (28)	**Q4** (30)
Q5 (3)	**Q6** (2)	**Q7** (4)	**Q8** (4)

Q1 (781)

x	–2	–1	3	4	6
P(X = x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

$\overline{X} = 2.3$

$-a + 6b = \frac{9}{10}$ ......(1)

$\sum P_i = \frac{1}{5} + a + \frac{1}{3} + \frac{1}{5} + b = 1$

$a + b = \frac{4}{15}$ ......(2)

From equation (1) and (2)

$a = \frac{1}{10},\ b = \frac{1}{6}$

$\sigma^2 = \Sigma p_i x_i^2 - (\overline{X})^2$

$\frac{1}{5}(4) + a(1) + \frac{1}{3}(9) + \frac{1}{5}(16) + b(36) - (2.3)^2$

$= \frac{4}{5} + a + 3 + \frac{16}{5} + 36\,b - (2.3)^2$

$= 4 + a + 3 + 36\,b - (2.3)^2$

$= 7 + a + 36\,b - (2.3)^2$

$= 7 + \frac{1}{10} + 6 - (2.3)^2$

$= 13 + \frac{1}{10} - \left(\frac{23}{10}\right)^2$

$= \frac{131}{10} - \left(\frac{23}{10}\right)^2$

$= \frac{1310-(23)^2}{100}$

$= \frac{1310-529}{100}$

$\sigma^2 = \frac{781}{100}$

$100\sigma^2 = 781$

Q2 (2)

Total ways of choosing square $= {}^{64}C_2$

$= \frac{64\times63}{2\times1} = 32 \times 63$

ways of choosing two squares having common side $= 2(7 \times 8) = 112$

Required probability $= \frac{112}{32 \times 63} = \frac{16}{32 \times 9} = \frac{1}{18}$.

Ans. (2)

Q3 (28)

$\mathrm{I}_1 =$ first unit is functioning

$\mathrm{I}_2 =$ second unit is functioning

$\mathrm{P}\left(\mathrm{I}_1\right) = 0.9, \mathrm{P}\left(\mathrm{I}_2\right) = 0.8$

$\mathrm{P}\left(\bar{\mathrm{I}}_1\right) = 0.1, \mathrm{P}\left(\bar{\mathrm{I}}_2\right) = 0.2$

$\mathrm{P} = \frac{0.8 \times 0.1}{0.1 \times 0.2 + 0.9 \times 0.2 + 0.1 \times 0.8} = \frac{8}{28}$

$98\mathrm{P} = \frac{8}{28} \times 98 = 28$

Q4 (30)

$\sum \mathrm{P}(\mathrm{X}) = 1 \Rightarrow \mathrm{k} + 2\mathrm{k} + 2\mathrm{k} + 3\mathrm{k} + \mathrm{k} = 1$

$\Rightarrow \mathrm{k} = \frac{1}{9}$

Now, $\mathrm{p} = \mathrm{P}\left(\frac{\mathrm{kX} < 4}{\mathrm{X} < 3}\right) = \frac{\mathrm{P}(\mathrm{X}=2)}{\mathrm{P}(\mathrm{X}<3)} = \frac{\frac{2\mathrm{k}}{9\mathrm{k}}}{\frac{\mathrm{k}}{9\mathrm{k}} + \frac{2\mathrm{k}}{9\mathrm{k}}} = \frac{2}{3}$

$\Rightarrow \mathrm{p} = \frac{2}{3}$

Now, $5\mathrm{p} = \lambda \mathrm{k}$

$\Rightarrow (5)\left(\frac{2}{3}\right) = \lambda(1/9)$

$\Rightarrow \lambda = 30$

Q5 (3)

C – I '0' Head

T T T $\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^3=\frac{1}{64}$

C – II '1' head

H T T $\left(\frac{3}{8}\right)\left(\frac{3}{8}\right)=\frac{9}{64}$

C – III '2' Head

H H T $\left(\frac{3}{8}\right)\left(\frac{3}{8}\right)=\frac{9}{64}$

C–IV '3' Heads

H H H $\left(\frac{1}{8}\right)\left(\frac{1}{8}\right)=\frac{1}{64}$

Total probability $=\frac{5}{16}$.

Q6 (2)

Probability of obtaining total sum $7=$ probability of getting opposite faces.

Probability of getting opposite faces

$=2\left[\left(\frac{1}{6}-x\right)\left(\frac{1}{6}+x\right)+\frac{1}{6}\times\frac{1}{6}+\frac{1}{6}\times\frac{1}{6}\right]$

$\Rightarrow 2\left[\left(\frac{1}{6}-x\right)\left(\frac{1}{6}+x\right)+\frac{1}{6}\times\frac{1}{6}+\frac{1}{6}\times\frac{1}{6}\right]=\frac{13}{96}$

(given)

$x=\frac{1}{8}$

Q7 (4)

$P(x\geq 5\mid x>2)=\frac{P(x\geq 5)}{P(x>2)}$

$\frac{\left(\frac{5}{6}\right)^4\cdot\frac{1}{6}+\left(\frac{5}{6}\right)^5\cdot\frac{1}{6}+\ldots\ldots+\infty}{\left(\frac{5}{6}\right)^2\cdot\frac{1}{6}+\left(\frac{5}{6}\right)^3\cdot\frac{1}{6}+\ldots\ldots+\infty}$

$$\frac{\dfrac{\left(\frac{5}{6}\right)^4 \cdot \frac{1}{6}}{1-\frac{5}{6}}}{\dfrac{\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}}{1-\frac{5}{6}}}=\left(\frac{5}{6}\right)^2=\frac{25}{36}$$

Q8 (4)

P(Exactly one of A or B)

$= \mathrm{P}(\mathrm{A} \cap \overline{\mathrm{B}}) + \mathrm{P}(\overline{\mathrm{A}} \cap \mathrm{B}) = \frac{5}{9}$

$= \mathrm{P}(\mathrm{A})\mathrm{P}(\overline{\mathrm{B}}) + \mathrm{P}(\overline{\mathrm{A}})\mathrm{P}(\mathrm{B}) = \frac{5}{9}$

$\Rightarrow \mathrm{P}(\mathrm{A})(1-\mathrm{P}(\mathrm{B})) + (1-\mathrm{P}(\mathrm{A}))\mathrm{P}(\mathrm{B}) = \frac{5}{9}$

$\Rightarrow \mathrm{p}(1-2\mathrm{p}) + (1-\mathrm{p})2\mathrm{p} = \frac{5}{9}$

$\Rightarrow 36\mathrm{p}^2 - 27\mathrm{p} + 5 = 0$

$\Rightarrow \mathrm{p} = \frac{1}{3}$ or $\frac{5}{12}$

$\mathrm{P}_{\max} = \frac{5}{12}$